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# 1 Object Diagram - Admin Context

UML Type: Object Diagram Category: Admin Source: docs/diagrams/objects/admin/admin\_objects.puml

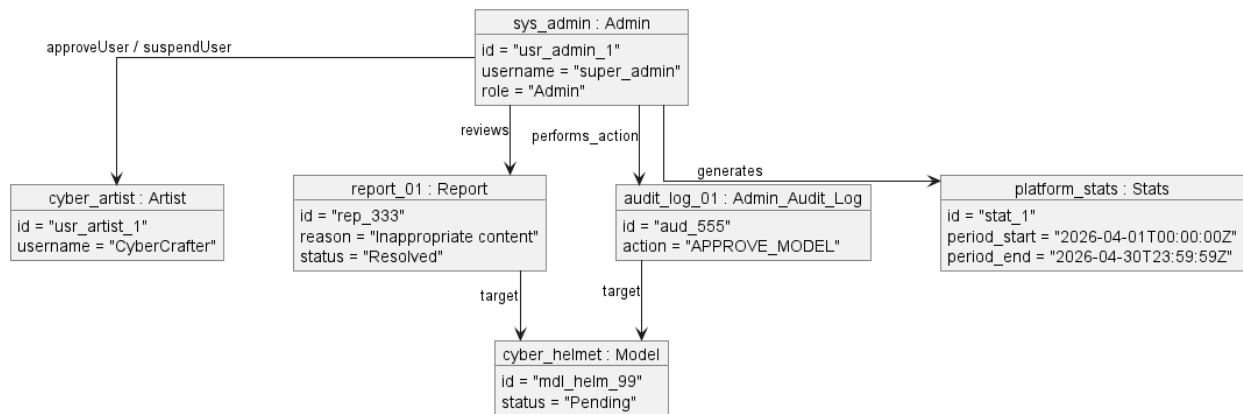


Figure 1: Object Diagram - Admin Context

## 1.1 Overview

This object diagram captures a runtime snapshot of the admin domain, showing how an Admin user interacts with models, reports, audit logs, and stats.

## 1.2 Objects

### 1.2.1 sys\_admin : Admin

The platform administrator instance.

| Attribute | Value         |
|-----------|---------------|
| id        | "usr_admin_1" |
| username  | "super_admin" |
| role      | "Admin"       |

### 1.2.2 cyber\_artist : Artist

An artist whose account and models are subject to admin actions.

| Attribute | Value          |
|-----------|----------------|
| id        | "usr_artist_1" |
| username  | "CyberCrafter" |

### 1.2.3 cyber\_helmet : Model

A model currently awaiting admin review.

| Attribute | Value                            |
|-----------|----------------------------------|
| id        | “mdl_helm_99”                    |
| status    | “Pending” (Model_Status.PENDING) |

### 1.2.4 report\_01 : Report

A resolved abuse report targeting the model.

| Attribute | Value                               |
|-----------|-------------------------------------|
| id        | “rep_333”                           |
| reason    | “Inappropriate content”             |
| status    | “Resolved” (Report_Status.REVIEWED) |

### 1.2.5 audit\_log\_01 : Admin\_Audit\_Log

An audit log entry created when the admin acted on the model.

| Attribute | Value                                        |
|-----------|----------------------------------------------|
| id        | “aud_555”                                    |
| action    | “APPROVE_MODEL” (Audit_Action.APPROVE_MODEL) |

### 1.2.6 platform\_stats : Stats

A periodic platform statistics snapshot generated by the admin.

| Attribute    | Value                  |
|--------------|------------------------|
| id           | “stat_1”               |
| period_start | “2026-04-01T00:00:00Z” |
| period_end   | “2026-04-30T23:59:59Z” |

## 1.3 Relationships

| From         | Relationship    | To           | Description                                 |
|--------------|-----------------|--------------|---------------------------------------------|
| sys_admin    | performs_action | audit_log_01 | Admin’s action is recorded in the audit log |
| audit_log_01 | target          | cyber_helmet | The model is the target of the audit action |

| From      | Relationship              | To             | Description                         |
|-----------|---------------------------|----------------|-------------------------------------|
| sys_admin | reviews                   | report_01      | Admin processes the pending report  |
| report_01 | target                    | cyber_helmet   | The report points to the model      |
| sys_admin | generates                 | platform_stats | Admin triggers stats aggregation    |
| sys_admin | approveUser / suspendUser | cyber_artist   | Admin manages artist account status |

## 1.4 System Behavior Notes

- When an admin approves a model, the system calls `model.update({ status: APPROVED })` and creates an `Admin_Audit_Log` entry. The diagram shows both the approval action (`audit_log action = APPROVE_MODEL`) and the model's earlier pending state.
- The `Report` record with `status = "Resolved"` reflects a Report that the admin has reviewed and closed. This maps to `Report_Status.REVIEWED` in the schema.
- The `platform_stats` object captures a full month period. The `data` field (not shown here) would contain a JSON structure with totals for orders, revenue, new users, and top models.
- Admin audit logs are append-only. The `admin_id` FK on `Admin_Audit_Log` connects back to the admin user, enabling the audit log viewer in the dashboard.

## 2 Object Diagram - Auth Context

**UML Type:** Object Diagram **Category:** Auth **Source:** docs/diagrams/objects/auth/auth\_objects.puml

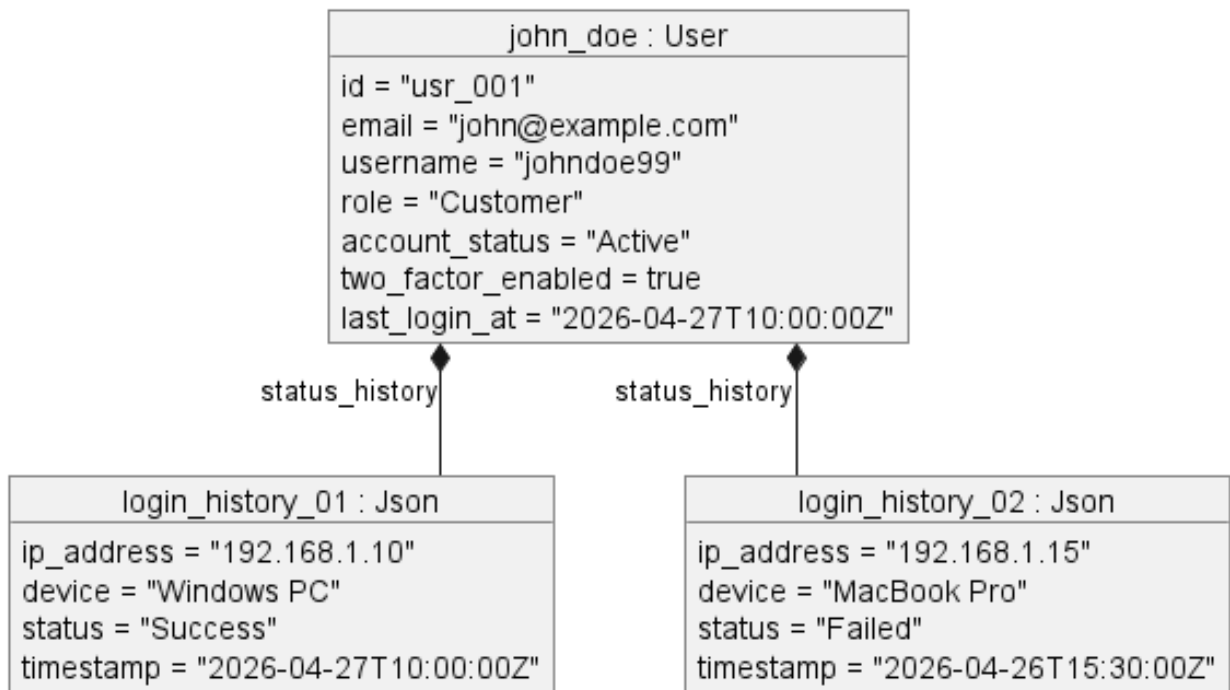


Figure 2: Object Diagram - Auth Context

---

## 2.1 Overview

This object diagram illustrates the runtime state of a `User` and their associated `status_history` JSON entries after two login attempts. It models the auth-context snapshot as it exists in the database.

---

## 2.2 Objects

### 2.2.1 `john_doe` : `User`

A concrete instance of the `User` entity representing an authenticated customer.

| Attribute                       | Value                                      |
|---------------------------------|--------------------------------------------|
| <code>id</code>                 | <code>"usr_001"</code>                     |
| <code>email</code>              | <code>"john@example.com"</code>            |
| <code>username</code>           | <code>"johndoe99"</code>                   |
| <code>role</code>               | <code>"Customer"</code>                    |
| <code>account_status</code>     | <code>"Active"</code> (APPROVED in schema) |
| <code>two_factor_enabled</code> | <code>true</code>                          |
| <code>last_login_at</code>      | <code>"2026-04-27T10:00:00Z"</code>        |

Note: `account_status = "Active"` in the diagram corresponds to `Account_Status.APPROVED` in the Prisma schema.

---

### 2.2.2 `login_history_01` : `Json`

A JSON entry embedded in the `status_history` array on the `User` record. Represents a successful login event.

| Attribute               | Value                               |
|-------------------------|-------------------------------------|
| <code>ip_address</code> | <code>"192.168.1.10"</code>         |
| <code>device</code>     | <code>"Windows PC"</code>           |
| <code>status</code>     | <code>"Success"</code>              |
| <code>timestamp</code>  | <code>"2026-04-27T10:00:00Z"</code> |

---

### 2.2.3 `login_history_02` : `Json`

A second embedded JSON entry representing a failed login attempt.

| Attribute               | Value                               |
|-------------------------|-------------------------------------|
| <code>ip_address</code> | <code>"192.168.1.15"</code>         |
| <code>device</code>     | <code>"MacBook Pro"</code>          |
| <code>status</code>     | <code>"Failed"</code>               |
| <code>timestamp</code>  | <code>"2026-04-26T15:30:00Z"</code> |

---

## 2.3 Relationships

| From            | Relationship                    | To                      | Description                                         |
|-----------------|---------------------------------|-------------------------|-----------------------------------------------------|
| john_doe (User) | status_history<br>(composition) | login_history_01 (Json) | First entry in the<br>status_history Json<br>array  |
| john_doe (User) | status_history<br>(composition) | login_history_02 (Json) | Second entry in the<br>status_history Json<br>array |

The composition notation indicates that `login_history_01` and `login_history_02` are owned by and cannot exist independently from `john_doe`. In practice, they are stored directly as JSON objects in the `status_history` column.

## 2.4 System Behavior Notes

- The `status_history` field is a `Json[]` column on the `User` table in Prisma. When an admin approves or rejects a user, a new JSON record is appended to this array.
- This diagram represents the snapshot after one successful session and one failed attempt, showing that the platform tracks session events in-place on the User record.
- The `last_login_at` field is updated by the `AuthService` on every successful login, as shown in the `sequence_local_login.puml` diagram.

# 3 Object Diagram - Catalog Context

**UML Type:** Object Diagram **Category:** Catalog **Source:** docs/diagrams/objects/catalog/catalog\_objects.puml

## 3.1 Overview

This object diagram represents the runtime state of the catalog domain. It shows a specific 3D model instance linked to its creator (Artist), its category, and its tags.

## 3.2 Objects

### 3.2.1 cyber\_artist : Artist

The creator of the featured model.

| Attribute | Value          |
|-----------|----------------|
| id        | "usr_artist_1" |
| username  | "CyberCrafter" |
| role      | "Artist"       |

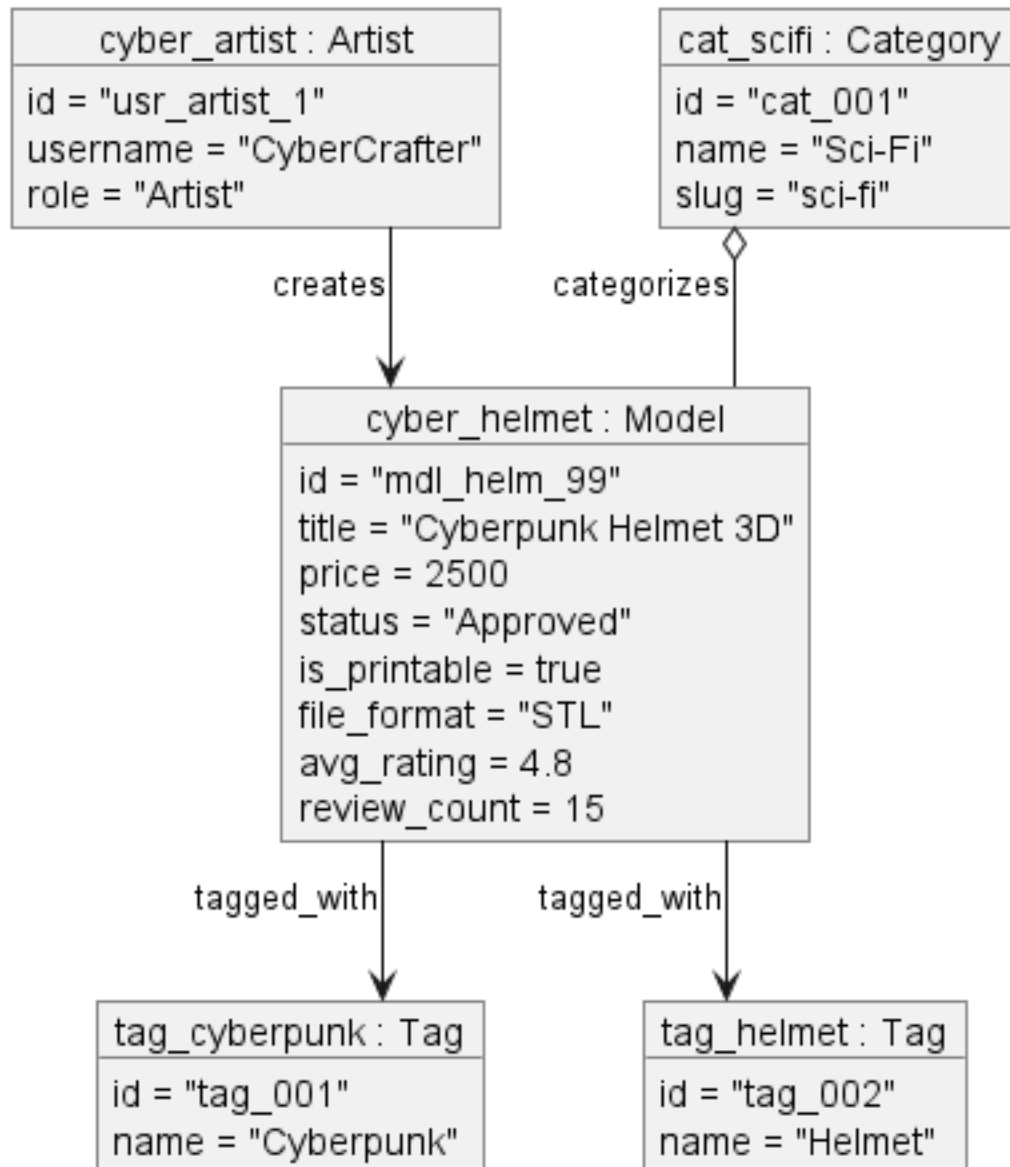


Figure 3: Object Diagram - Catalog Context

### 3.2.2 cat\_scifi : Category

The category the model belongs to.

| Attribute | Value     |
|-----------|-----------|
| id        | "cat_001" |
| name      | "Sci-Fi"  |
| slug      | "sci-fi"  |

---

### 3.2.3 tag\_cyberpunk : Tag

A descriptive tag applied to the model.

| Attribute | Value       |
|-----------|-------------|
| id        | "tag_001"   |
| name      | "Cyberpunk" |

---

### 3.2.4 tag\_helmet : Tag

A second descriptive tag applied to the model.

| Attribute | Value     |
|-----------|-----------|
| id        | "tag_002" |
| name      | "Helmet"  |

---

### 3.2.5 cyber\_helmet : Model

The 3D model at the centre of this snapshot.

| Attribute    | Value                              |
|--------------|------------------------------------|
| id           | "mdl_helm_99"                      |
| title        | "Cyberpunk Helmet 3D"              |
| price        | 2500 (IDR)                         |
| status       | "Approved" (Model_Status.APPROVED) |
| is_printable | true                               |
| file_format  | "STL"                              |
| avg_rating   | 4.8                                |
| review_count | 15                                 |

---

Note: The file\_format field shows "STL" in this object diagram. The Prisma schema defaults to "glb" for uploaded models. This is technically inconsistent - the system accepts GLB/GLTF files via the upload flow. STL would need explicit handling.

---

### 3.3 Relationships

| From         | Relationship | To            |
|--------------|--------------|---------------|
| cyber_artist | creates      | cyber_helmet  |
| cat_scifi    | categorizes  | cyber_helmet  |
| cyber_helmet | tagged_with  | tag_cyberpunk |
| cyber_helmet | tagged_with  | tag_helmet    |

The `categorizes` link is represented with an aggregation (open diamond) notation indicating that a Category can exist without any models. The `creates` link is a directed association from Artist to Model. Tags are connected via a many-to-many implicit join table managed by Prisma.

### 3.4 System Behavior Notes

- The `status = "Approved"` state means this model is publicly visible in the catalog to non-admin users.
- The `avg_rating` and `review_count` are denormalized fields updated after each review submission by aggregating the `Review` table.
- The `is_printable = true` flag makes this model available for selection in the Print Order flow.

## 4 Object Diagram - Commerce Context

**UML Type:** Object Diagram **Category:** Commerce **Source:** docs/diagrams/objects/commerce/commerce\_object

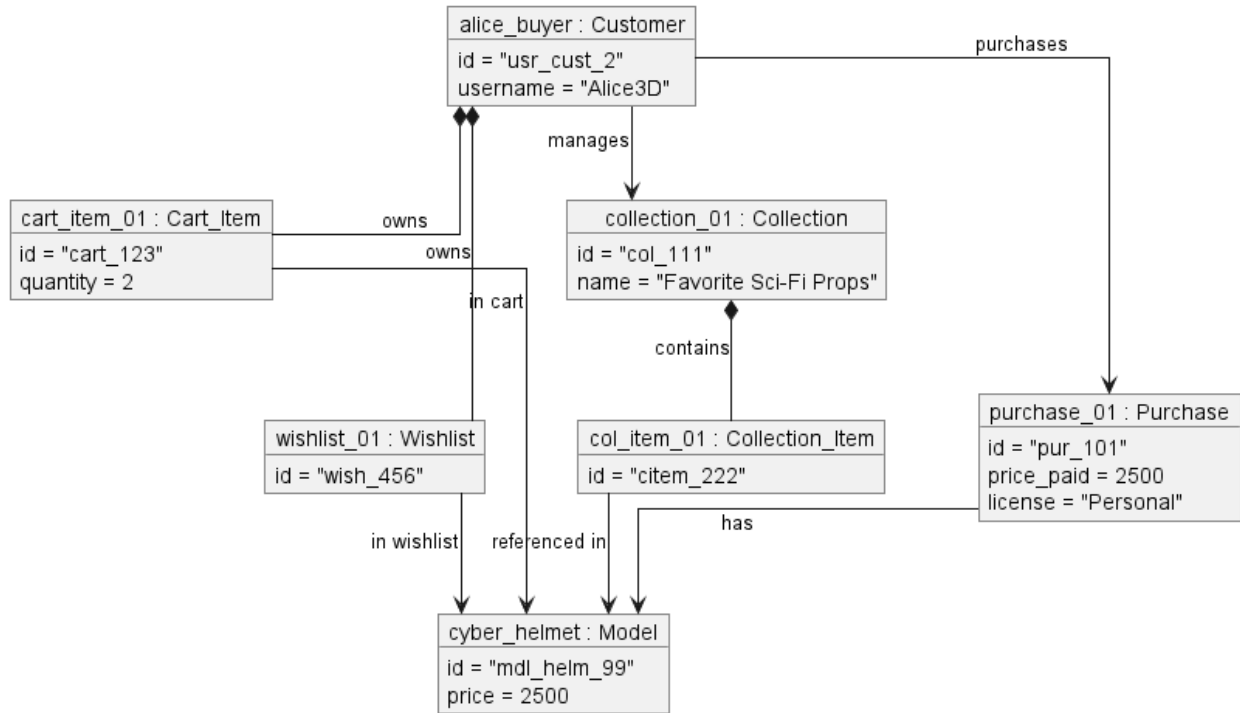


Figure 4: Object Diagram - Commerce Context

## 4.1 Overview

This object diagram shows a runtime snapshot of the commerce domain. It models a customer's cart, wishlist, collection, and completed purchase for a single model.

---

## 4.2 Objects

### 4.2.1 `alice_buyer` : `Customer`

The purchasing user.

---

| Attribute | Value        |
|-----------|--------------|
| id        | "usr_cust_2" |
| username  | "Alice3D"    |

---

### 4.2.2 `cyber_helmet` : `Model`

The 3D model in question.

---

| Attribute | Value         |
|-----------|---------------|
| id        | "mdl_helm_99" |
| price     | 2500 (IDR)    |

---

### 4.2.3 `cart_item_01` : `Cart_Item`

A cart entry for the model.

---

| Attribute | Value      |
|-----------|------------|
| id        | "cart_123" |
| quantity  | 2          |

---

Note: The diagram shows quantity = 2. In the real system, the cart normalization logic forces quantity = 1 for all digital models (`cart_Item.updateMany` sets quantity to 1 on every cart fetch). This object is a conceptual representation.

---

### 4.2.4 `wishlist_01` : `Wishlist`

A wishlist entry for the model.

---

| Attribute | Value      |
|-----------|------------|
| id        | "wish_456" |

---

#### 4.2.5 collection\_01 : Collection

A named collection owned by the customer.

| Attribute | Value                   |
|-----------|-------------------------|
| id        | "col_111"               |
| name      | "Favorite Sci-Fi Props" |

#### 4.2.6 col\_item\_01 : Collection\_Item

The join record linking the collection to the model.

| Attribute | Value       |
|-----------|-------------|
| id        | "citem_222" |

#### 4.2.7 purchase\_01 : Purchase

The confirmed purchase record created after payment success.

| Attribute  | Value                                  |
|------------|----------------------------------------|
| id         | "pur_101"                              |
| price_paid | 2500 (IDR)                             |
| license    | "Personal" (License_Type.PERSONAL_USE) |

### 4.3 Relationships

| From          | Relationship           | To            | Description                           |
|---------------|------------------------|---------------|---------------------------------------|
| alice_buyer   | owns (composition)     | cart_item_01  | Cart item belongs to customer         |
| cart_item_01  | in cart                | cyber_helmet  | Model referenced in cart              |
| alice_buyer   | owns (composition)     | wishlist_01   | Wishlist entry belongs to customer    |
| wishlist_01   | in wishlist            | cyber_helmet  | Model saved to wishlist               |
| alice_buyer   | manages                | collection_01 | Customer owns the collection          |
| collection_01 | contains (composition) | col_item_01   | Collection item belongs to collection |
| col_item_01   | referenced in          | cyber_helmet  | Model referenced in collection        |
| alice_buyer   | purchases              | purchase_01   | Confirmed ownership record            |
| purchase_01   | has                    | cyber_helmet  | Purchase links to the model           |

## 4.4 System Behavior Notes

- In the real system, a user can simultaneously have a model in their wishlist, cart, collection, and purchase records. These are four independent tables in Prisma.
- The Purchase record is only created by the Midtrans webhook handler after `transaction_status = "settlement"`. A cart item alone does not mean the user owns the model.
- The Collection\_Item is deleted via CASCADE if either the parent Collection or the referenced Model is deleted.
- The Wishlist entry is similarly CASCADE-deleted if the model is deleted.

## 5 Object Diagram - Complete System Snapshot

**UML Type:** Object Diagram **Category:** Uncategorized (Complete System Overview) **Source:** docs/diagrams/objects/complete\_system\_object\_diagram.puml

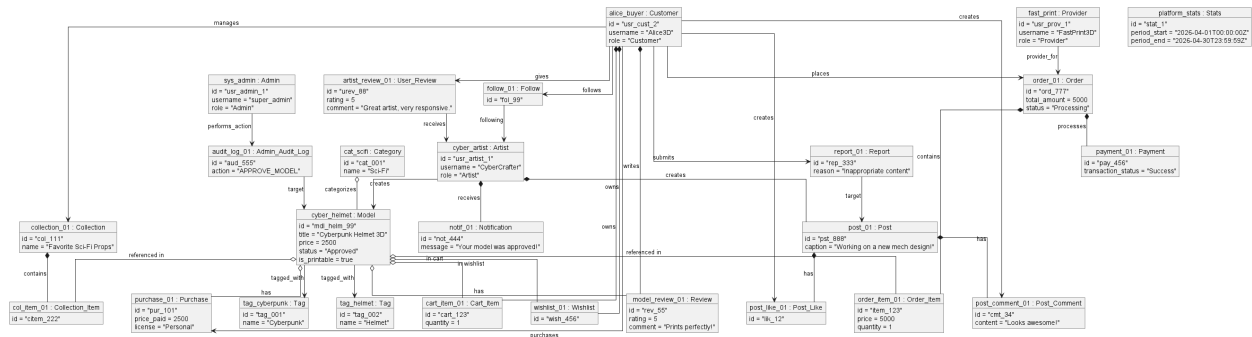


Figure 5: Object Diagram - Complete System Snapshot

### 5.1 Overview

This is the master object diagram for the entire 3Dex platform. It instantiates one representative object from every major domain and connects them to illustrate how all parts of the system interact in a single coherent runtime snapshot.

### 5.2 Objects by Domain

#### 5.2.1 Users

| Object Name  | Type     | Key Attributes                                                  |
|--------------|----------|-----------------------------------------------------------------|
| sys_admin    | Admin    | id="usr_admin_1",<br>username="super_admin",<br>role="Admin"    |
| cyber_artist | Artist   | id="usr_artist_1",<br>username="CyberCrafter",<br>role="Artist" |
| alice_buyer  | Customer | id="usr_cust_2",<br>username="Alice3D",<br>role="Customer"      |

| Object Name | Type     | Key Attributes                                                 |
|-------------|----------|----------------------------------------------------------------|
| fast_print  | Provider | id="usr_prov_1",<br>username="FastPrint3D",<br>role="Provider" |

### 5.2.2 Catalog and Models

| Object Name   | Type     | Key Attributes                                                                                           |
|---------------|----------|----------------------------------------------------------------------------------------------------------|
| cat_scifi     | Category | id="cat_001", name="Sci-Fi"                                                                              |
| tag_cyberpunk | Tag      | id="tag_001",<br>name="Cyberpunk"                                                                        |
| tag_helmet    | Tag      | id="tag_002", name="Helmet"                                                                              |
| cyber_helmet  | Model    | id="mdl_helm_99",<br>title="Cyberpunk Helmet 3D",<br>price=2500, status="Approved",<br>is_printable=true |

### 5.2.3 Shopping and Orders

| Object Name   | Type       | Key Attributes                                             |
|---------------|------------|------------------------------------------------------------|
| purchase_01   | Purchase   | id="pur_101", price_paid=2500,<br>license="Personal"       |
| cart_item_01  | Cart_Item  | id="cart_123", quantity=1                                  |
| wishlist_01   | Wishlist   | id="wish_456"                                              |
| order_01      | Order      | id="ord_777",<br>total_amount=5000,<br>status="Processing" |
| order_item_01 | Order_Item | id="item_123", price=5000,<br>quantity=1                   |
| payment_01    | Payment    | id="pay_456",<br>transaction_status="Success"              |

### 5.2.4 Social and Community

| Object Name      | Type         | Key Attributes                                                         |
|------------------|--------------|------------------------------------------------------------------------|
| model_review_01  | Review       | id="rev_55", rating=5,<br>comment="Prints perfectly!"                  |
| artist_review_01 | User_Review  | id="urev_88", rating=5,<br>comment="Great artist, very<br>responsive." |
| post_01          | Post         | id="pst_888",<br>caption="Working on a new<br>mech design!"            |
| post_like_01     | Post_Like    | id="lik_12"                                                            |
| post_comment_01  | Post_Comment | id="cmt_34", content="Looks<br>awesome!"                               |
| follow_01        | Follow       | id="fol_99"                                                            |

| Object Name   | Type            | Key Attributes                             |
|---------------|-----------------|--------------------------------------------|
| collection_01 | Collection      | id="col_111", name="Favorite Sci-Fi Props" |
| col_item_01   | Collection_Item | id="citem_222"                             |

### 5.2.5 System and Admin

| Object Name    | Type            | Key Attributes                                                        |
|----------------|-----------------|-----------------------------------------------------------------------|
| report_01      | Report          | id="rep_333",<br>reason="Inappropriate content"                       |
| notif_01       | Notification    | id="not_444", message="Your model was approved!"                      |
| audit_log_01   | Admin_Audit_Log | id="aud_555",<br>action="APPROVE_MODEL"                               |
| platform_stats | Stats           | id="stat_1",<br>period_start="2026-04-01",<br>period_end="2026-04-30" |

## 5.3 Relationships

### 5.3.1 Catalog and Model Relationships

| From         | Relationship | To            |
|--------------|--------------|---------------|
| cyber_artist | creates      | cyber_helmet  |
| cat_scifi    | categorizes  | cyber_helmet  |
| cyber_helmet | tagged_with  | tag_cyberpunk |
| cyber_helmet | tagged_with  | tag_helmet    |

### 5.3.2 Commerce Relationships

| From          | Relationship            | To            |
|---------------|-------------------------|---------------|
| alice_buyer   | purchases               | purchase_01   |
| purchase_01   | has                     | cyber_helmet  |
| alice_buyer   | owns (composition)      | cart_item_01  |
| cart_item_01  | in cart                 | cyber_helmet  |
| alice_buyer   | owns (composition)      | wishlist_01   |
| wishlist_01   | in wishlist             | cyber_helmet  |
| alice_buyer   | places                  | order_01      |
| fast_print    | provider_for            | order_01      |
| order_01      | contains (composition)  | order_item_01 |
| order_item_01 | referenced in           | cyber_helmet  |
| order_01      | processes (composition) | payment_01    |

### 5.3.3 Social Relationships

| From             | Relationship           | To               |
|------------------|------------------------|------------------|
| alice_buyer      | writes (composition)   | model_review_01  |
| model_review_01  | has                    | cyber_helmet     |
| alice_buyer      | gives                  | artist_review_01 |
| artist_review_01 | receives               | cyber_artist     |
| cyber_artist     | creates (composition)  | post_01          |
| alice_buyer      | creates                | post_like_01     |
| alice_buyer      | creates                | post_comment_01  |
| post_01          | has (composition)      | post_like_01     |
| post_01          | has (composition)      | post_comment_01  |
| alice_buyer      | follows                | follow_01        |
| follow_01        | following              | cyber_artist     |
| alice_buyer      | manages                | collection_01    |
| collection_01    | contains (composition) | col_item_01      |
| col_item_01      | referenced in          | cyber_helmet     |

#### 5.3.4 System and Admin Relationships

| From         | Relationship           | To           |
|--------------|------------------------|--------------|
| alice_buyer  | submits                | report_01    |
| report_01    | target                 | post_01      |
| cyber_artist | receives (composition) | notif_01     |
| sys_admin    | performs_action        | audit_log_01 |
| audit_log_01 | target                 | cyber_helmet |

### 5.4 System Behavior Notes

- This diagram is the union of all sub-category object diagrams. It demonstrates the full graph of dependencies in a single user journey where Alice browses, purchases, and reviews a model, while the artist posts content, the provider fulfills a print job, and the admin oversees everything.
- All `id` values are representative slugs, not real UUIDs, for readability.
- The `status = "Processing"` on `order_01` is a conceptual simplification. In the actual schema, this maps to intermediate states tracked by `Order_Item.print_status` rather than the `Order.status` field (which uses PENDING/PAID/FAILED/CANCELLED).

## 6 Object Diagram - Notifications Context

**UML Type:** Object Diagram **Category:** Notifications **Source:** docs/diagrams/objects/notifications/notifications

### 6.1 Overview

This object diagram shows a snapshot of the notifications domain, modeling two distinct notification records delivered to a single Artist user.

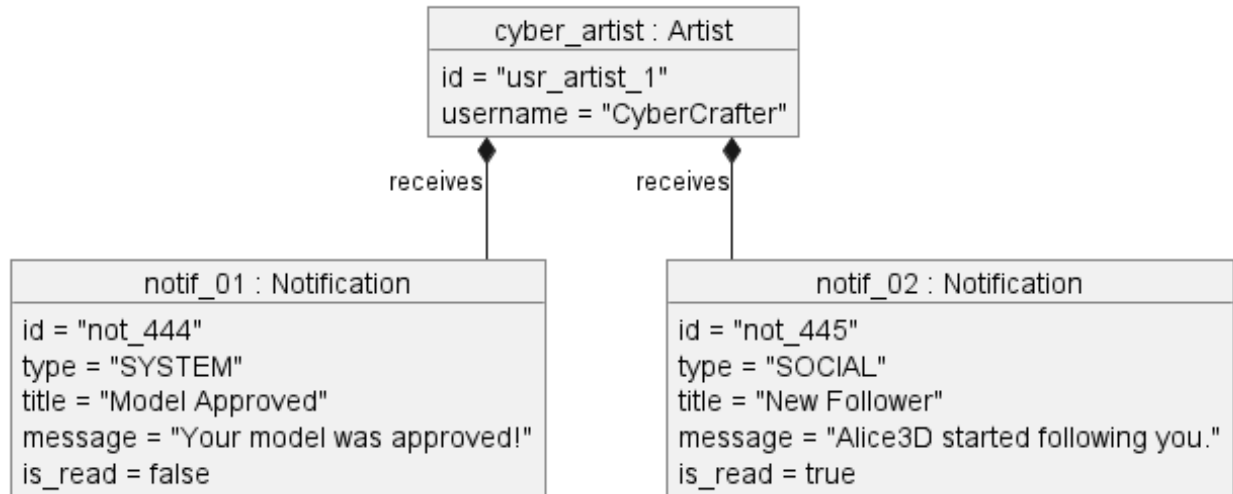


Figure 6: Object Diagram - Notifications Context

## 6.2 Objects

### 6.2.1 cyber\_artist : Artist

The notification recipient.

| Attribute | Value          |
|-----------|----------------|
| id        | "usr_artist_1" |
| username  | "CyberCrafter" |

### 6.2.2 notif\_01 : Notification

A system notification triggered by an admin action.

| Attribute | Value                      |
|-----------|----------------------------|
| id        | "not_444"                  |
| type      | "SYSTEM"                   |
| title     | "Model Approved"           |
| message   | "Your model was approved!" |
| is_read   | false                      |

### 6.2.3 notif\_02 : Notification

A social notification triggered by a follow action.

| Attribute | Value          |
|-----------|----------------|
| id        | "not_445"      |
| type      | "SOCIAL"       |
| title     | "New Follower" |

---

| Attribute | Value                            |
|-----------|----------------------------------|
| message   | “Alice3D started following you.” |
| is_read   | true                             |

---

### 6.3 Relationships

---

| From         | Relationship           | To       | Description                            |
|--------------|------------------------|----------|----------------------------------------|
| cyber_artist | receives (composition) | notif_01 | System notification<br>owned by artist |
| cyber_artist | receives (composition) | notif_02 | Social notification<br>owned by artist |

---

### 6.4 System Behavior Notes

- The **type** field is a free-form string in the schema. The values “SYSTEM” and “SOCIAL” are application-level conventions used by the `NotificationService`.
- The **is\_read** = **false** on `notif_01` means it will appear highlighted in the notification panel and contribute to the unread badge count.
- Notifications are created by `NotificationService.create_notification()` and dispatched from multiple points in the application: the payment webhook, the follow handler, and the print job lifecycle manager.
- The **data** field (not shown here) can carry optional JSON metadata, for example a model ID or post ID that the notification relates to, enabling deep-link navigation.

## 7 Object Diagram - Print Context

**UML Type:** Object Diagram **Category:** Print **Source:** docs/diagrams/objects/print/print\_objects.puml

---

### 7.1 Overview

This object diagram captures a runtime snapshot of the print job domain, modeling a full print order lifecycle from customer submission through payment. It shows the relationships between the customer, provider, order, items, configuration, payment, and model.

---

### 7.2 Objects

#### 7.2.1 alice\_buyer : Customer

The customer who submitted the print order.

---

| Attribute | Value        |
|-----------|--------------|
| id        | “usr_cust_2” |
| username  | “Alice3D”    |

---

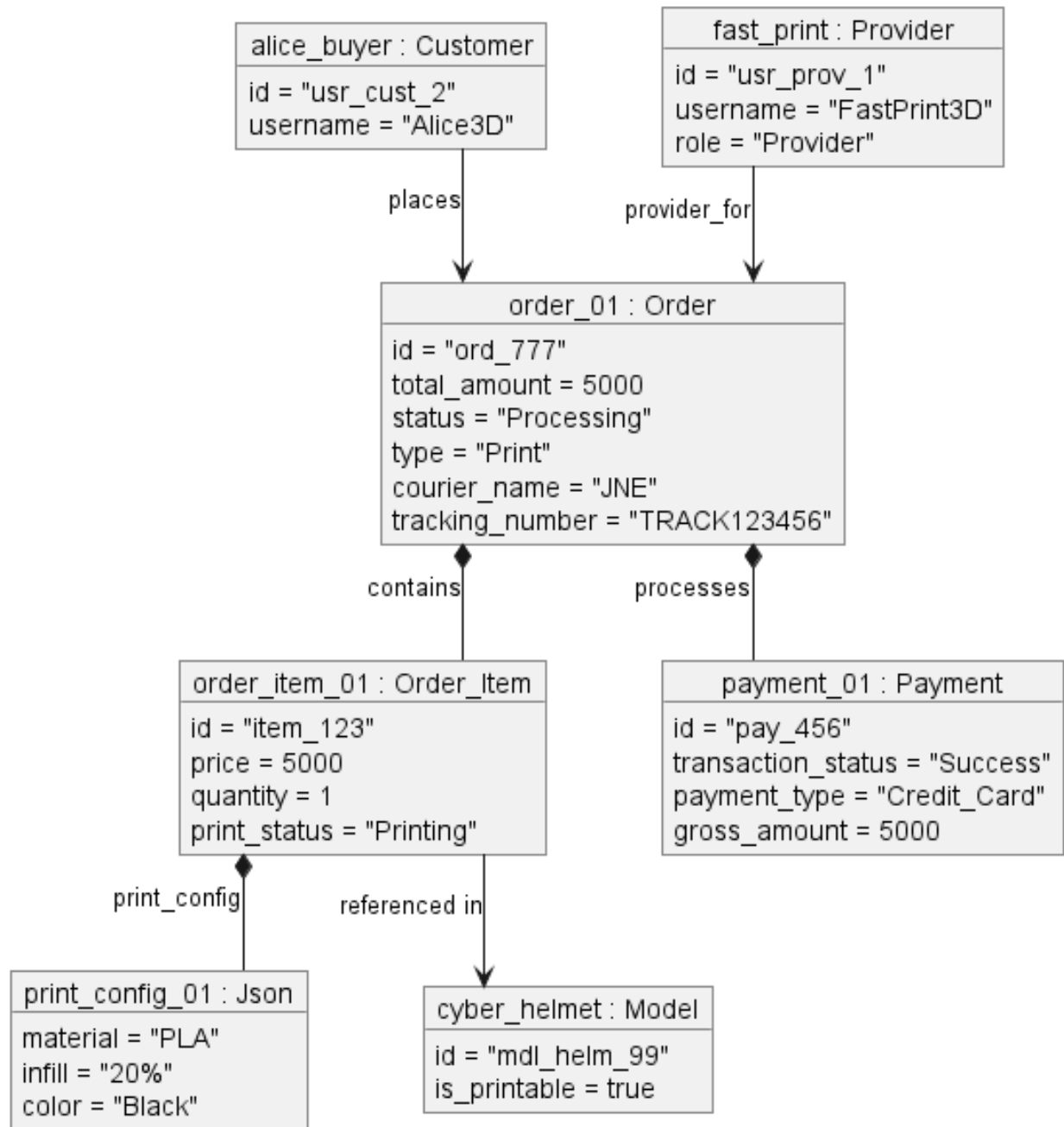


Figure 7: Object Diagram - Print Context

### 7.2.2 fast\_print : Provider

The approved 3D printing service provider assigned to fulfill the job.

| Attribute | Value         |
|-----------|---------------|
| id        | “usr_prov_1”  |
| username  | “FastPrint3D” |
| role      | “Provider”    |

### 7.2.3 order\_01 : Order

The print job order record.

| Attribute       | Value                                               |
|-----------------|-----------------------------------------------------|
| id              | “ord_777”                                           |
| total_amount    | 5000 (IDR)                                          |
| status          | “Processing” (Order_Status.PENDING or custom state) |
| type            | “Print” (Order_Type.PRINT_JOB)                      |
| courier_name    | “JNE”                                               |
| tracking_number | “TRACK123456”                                       |

Note: The diagram shows **status** = “Processing”. The schema does not have a “Processing” state in **Order\_Status**. The actual valid values are PENDING, PAID, FAILED, and CANCELLED. Processing state is tracked at the **Order\_Item** level via the **print\_status** field (Print\_Status enum: PENDING, ACCEPTED, PROCESSING, SHIPPED, DELIVERED, CANCELLED).

### 7.2.4 order\_item\_01 : Order\_Item

A single line item in the print order.

| Attribute    | Value                                |
|--------------|--------------------------------------|
| id           | “item_123”                           |
| price        | 5000 (IDR)                           |
| quantity     | 1                                    |
| print_status | “Printing” (Print_Status.PROCESSING) |

### 7.2.5 print\_config\_01 : Json

The print configuration JSON embedded in the **order\_item\_01** record.

| Attribute | Value   |
|-----------|---------|
| material  | “PLA”   |
| infill    | “20%”   |
| color     | “Black” |

### 7.2.6 payment\_01 : Payment

The successful Midtrans payment record for this order.

| Attribute          | Value                                   |
|--------------------|-----------------------------------------|
| id                 | “pay_456”                               |
| transaction_status | “Success” (Midtrans “settlement” value) |
| payment_type       | “Credit_Card”                           |
| gross_amount       | 5000 (IDR)                              |

### 7.2.7 cyber\_helmet : Model

The model being printed.

| Attribute    | Value         |
|--------------|---------------|
| id           | “mdl_helm_99” |
| is_printable | true          |

## 7.3 Relationships

| From          | Relationship               | To              | Description                            |
|---------------|----------------------------|-----------------|----------------------------------------|
| alice_buyer   | places                     | order_01        | Customer creates the order             |
| fast_print    | provider_for               | order_01        | Provider is assigned to the order      |
| order_01      | contains (composition)     | order_item_01   | Item is part of the order              |
| order_item_01 | print_config (composition) | print_config_01 | Config JSON embedded in item           |
| order_item_01 | referenced in              | cyber_helmet    | Item points to the model being printed |
| order_01      | processes (composition)    | payment_01      | Payment record linked to the order     |

## 7.4 System Behavior Notes

- Print orders use `Order_Type.PRINT_JOB`. The `provider_id` foreign key on `Order` is set at creation time when the customer selects a provider.
- Print configuration (material, infill, color, scale) is stored in the `print_config` JSON column on each `Order_Item`. The provider reads this to understand what to produce.
- The `print_status` field on `Order_Item` tracks the physical lifecycle of each item independently (PENDING -> ACCEPTED -> SHIPPED -> DELIVERED or CANCELLED).
- The payment for a print job is separate from digital asset payments. Print jobs may use a different billing arrangement since the `payment_01` shown here has a nonzero `gross_amount`.

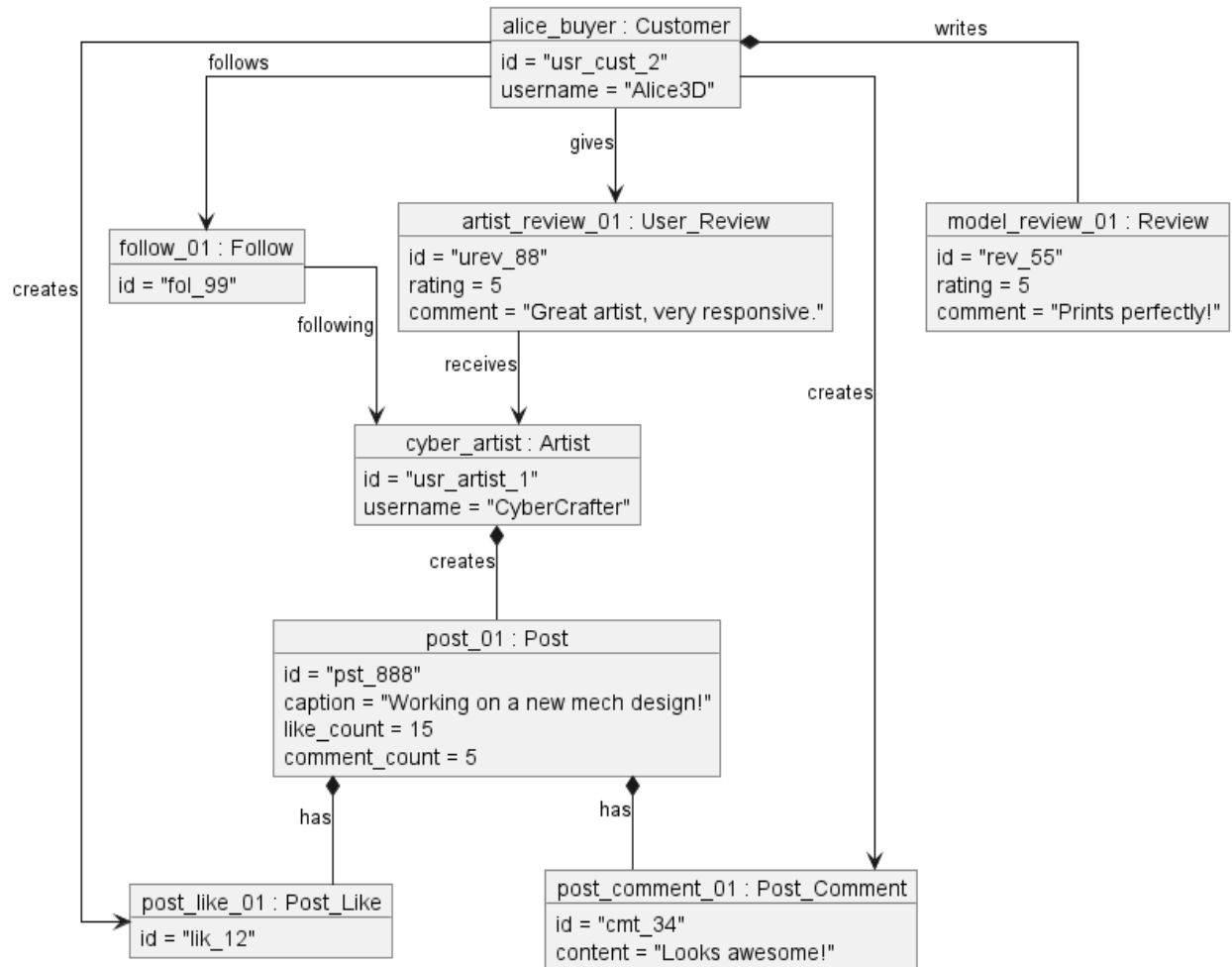


Figure 8: Object Diagram - Social Context

## 8 Object Diagram - Social Context

**UML Type:** Object Diagram **Category:** Social **Source:** docs/diagrams/objects/social/social\_objects.puml

---

### 8.1 Overview

This object diagram models the social domain at runtime. It shows how an Artist and a Customer interact through posts, likes, comments, follows, and reviews.

---

### 8.2 Objects

#### 8.2.1 cyber\_artist : Artist

The creator being followed and reviewed.

---

| Attribute | Value          |
|-----------|----------------|
| id        | "usr_artist_1" |
| username  | "CyberCrafter" |

---

#### 8.2.2 alice\_buyer : Customer

The user interacting with the artist's content.

---

| Attribute | Value        |
|-----------|--------------|
| id        | "usr_cust_2" |
| username  | "Alice3D"    |

---

#### 8.2.3 post\_01 : Post

A community post published by the artist.

---

| Attribute     | Value                           |
|---------------|---------------------------------|
| id            | "pst_888"                       |
| caption       | "Working on a new mech design!" |
| like_count    | 15                              |
| comment_count | 5                               |

---

#### 8.2.4 post\_like\_01 : Post\_Like

A like record created by Alice on the post.

---

| Attribute | Value    |
|-----------|----------|
| id        | “lik_12” |

---

### 8.2.5 post\_comment\_01 : Post\_Comment

A comment left by Alice on the post.

---

| Attribute | Value            |
|-----------|------------------|
| id        | “cmt_34”         |
| content   | “Looks awesome!” |

---

### 8.2.6 follow\_01 : Follow

A directed follow edge from Alice to the artist.

---

| Attribute | Value    |
|-----------|----------|
| id        | “fol_99” |

---

### 8.2.7 model\_review\_01 : Review

A model review written by Alice for one of the artist’s models.

---

| Attribute | Value               |
|-----------|---------------------|
| id        | “rev_55”            |
| rating    | 5                   |
| comment   | “Prints perfectly!” |

---

### 8.2.8 artist\_review\_01 : User\_Review

A user-level review of the artist as a seller, written by Alice.

---

| Attribute | Value                            |
|-----------|----------------------------------|
| id        | “urev_88”                        |
| rating    | 5                                |
| comment   | “Great artist, very responsive.” |

---

## 8.3 Relationships

| From             | Relationship          | To               | Description                     |
|------------------|-----------------------|------------------|---------------------------------|
| cyber_artist     | creates (composition) | post_01          | Artist owns the post            |
| alice_buyer      | creates               | post_like_01     | Alice created the like          |
| alice_buyer      | creates               | post_comment_01  | Alice created the comment       |
| post_01          | has (composition)     | post_like_01     | Like is part of the post        |
| post_01          | has (composition)     | post_comment_01  | Comment is part of the post     |
| alice_buyer      | follows               | follow_01        | Alice initiated the follow      |
| follow_01        | following             | cyber_artist     | The follow points to the artist |
| alice_buyer      | writes (composition)  | model_review_01  | Review owned by Alice           |
| alice_buyer      | gives                 | artist_review_01 | User review given by Alice      |
| artist_review_01 | receives              | cyber_artist     | Artist receives the user review |

## 8.4 System Behavior Notes

- **like\_count** and **comment\_count** on **Post** are denormalized counters. They are incremented/decremented atomically in the database alongside the create/delete of **Post\_Like** and **Post\_Comment** records.
- **Review** (model review) and **User\_Review** (person review) are two separate tables. A **Review** requires a prior **Purchase**, while a **User\_Review** requires either a purchase from the artist or a completed print job from the provider.
- The **Follow** table is directional: **follower\_id** is Alice, **following\_id** is the artist. When Alice follows the artist, a **Notification** is created for the artist.
- Post creation is restricted to users with **role** = **ARTIST** or **role** = **PROVIDER**. Customers cannot create posts.